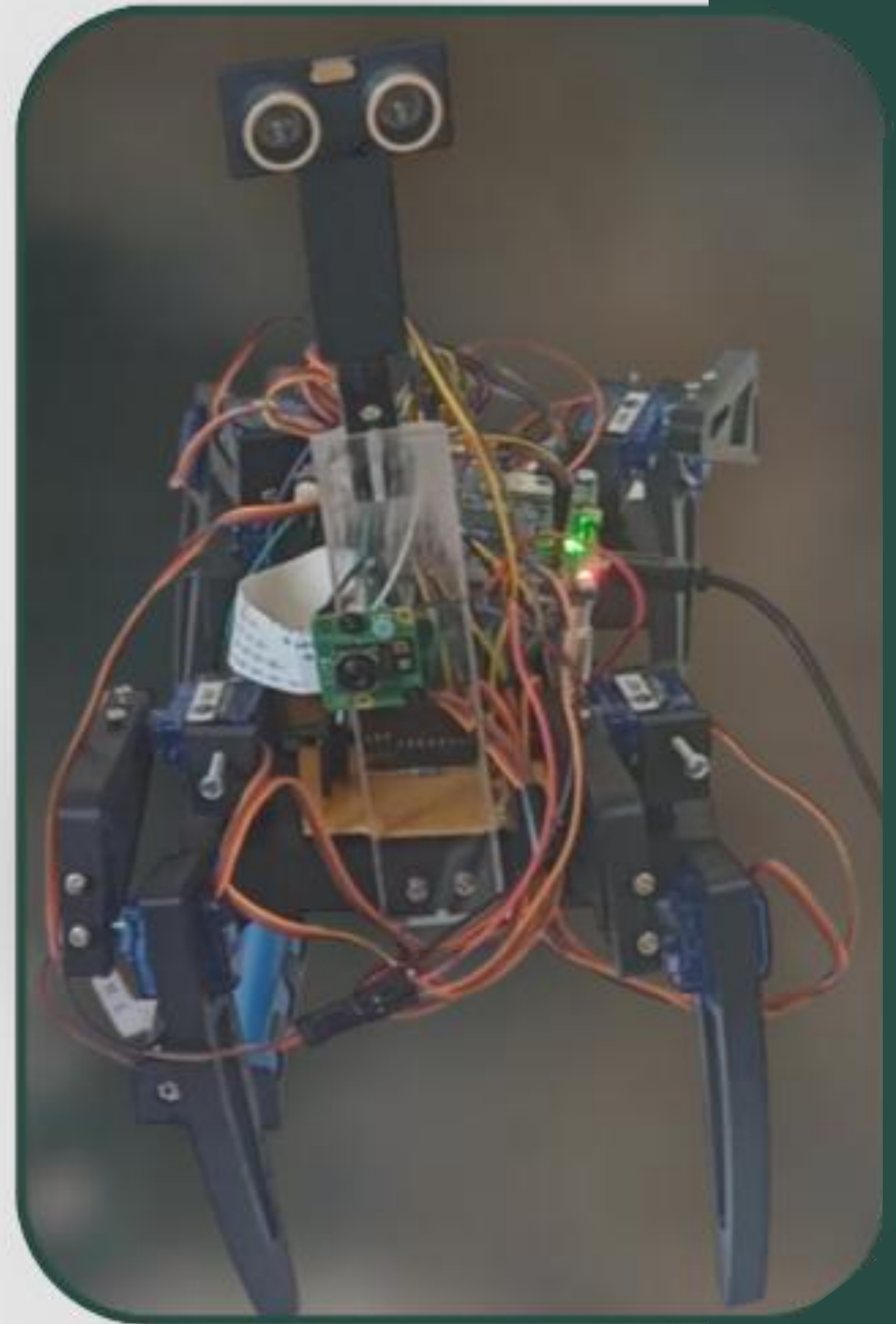


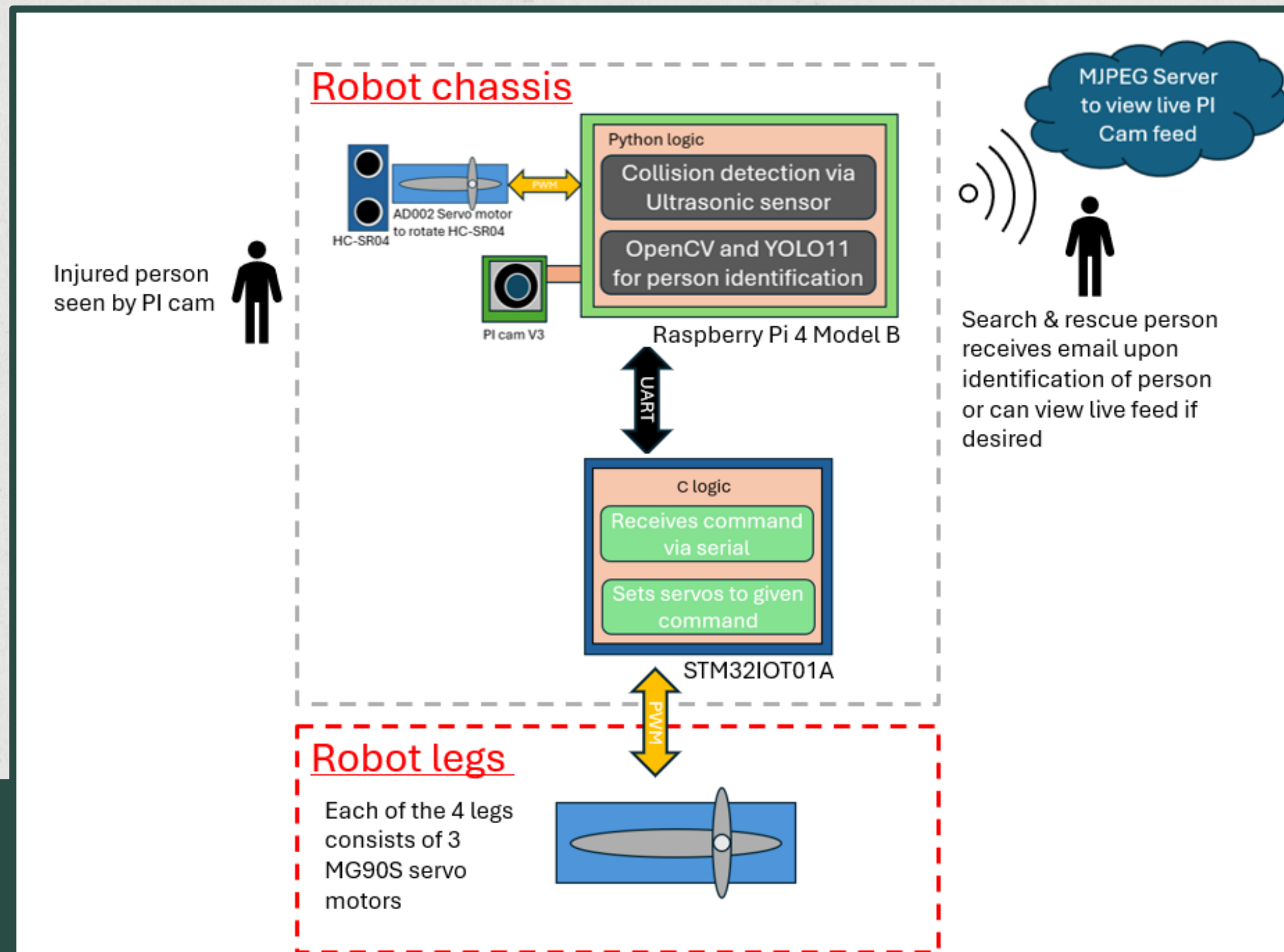
Search & Rescue Quadruped

Nathan Ferry G00413711

Final year project demonstration



System Architecture



YOLO 11

YOLO Object Detection Stream



Overall architecture

Residual Blocks

Boundry box regression

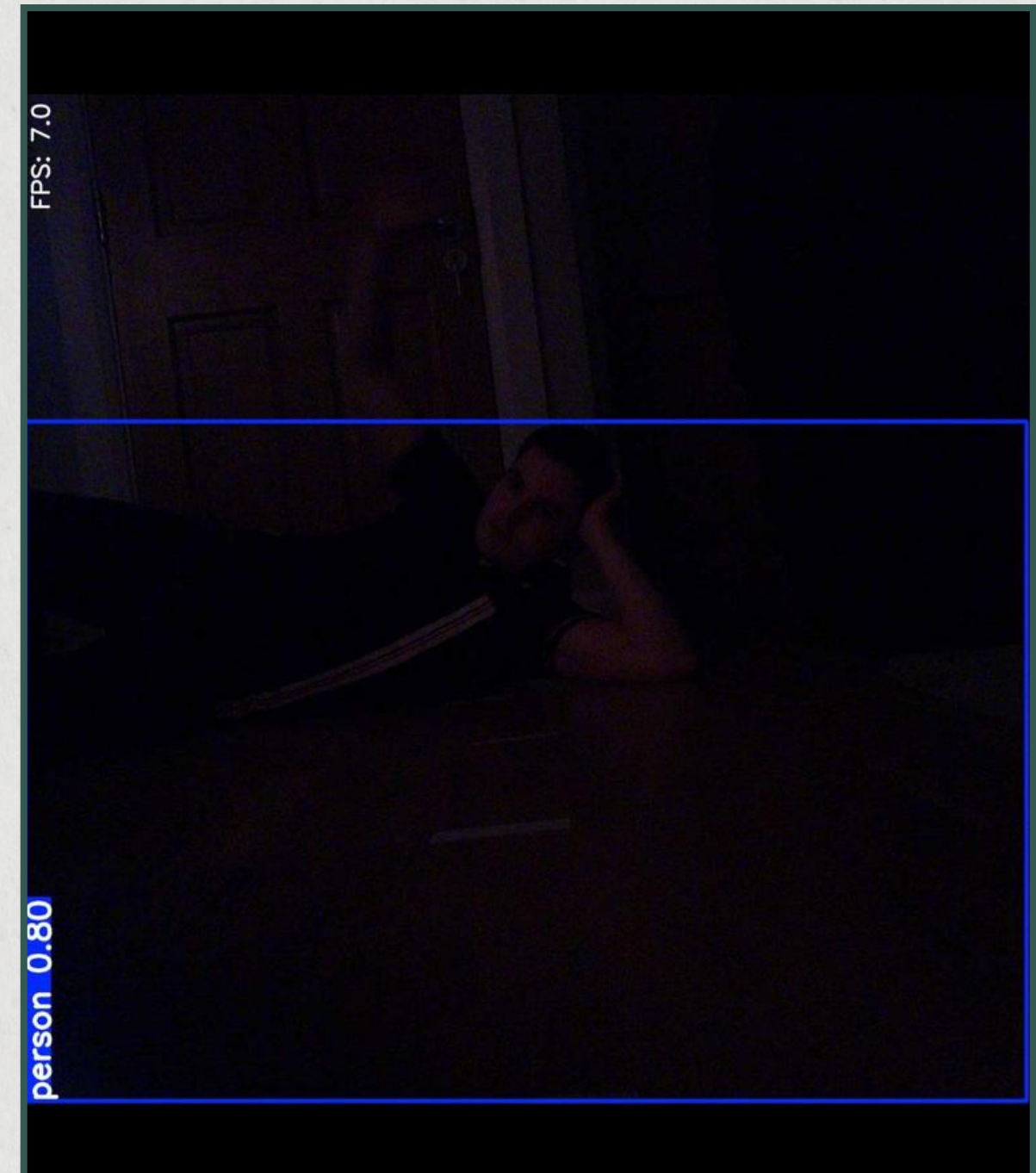
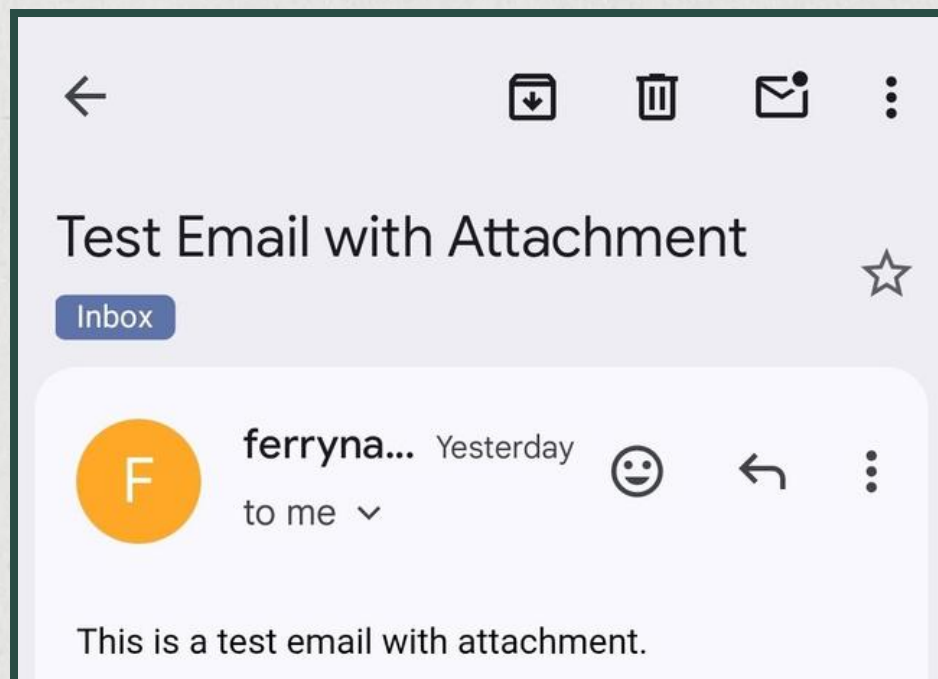
Intersection over union (IOU)

Non-max suppression (NMS)

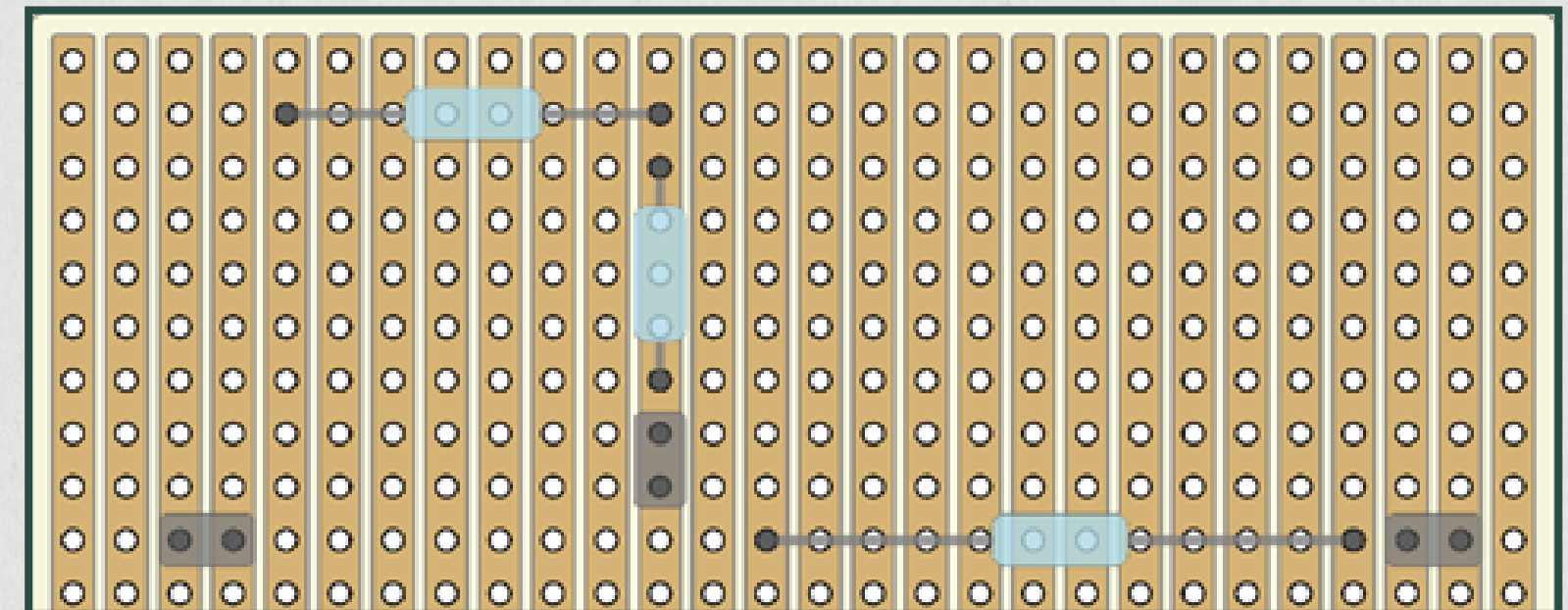
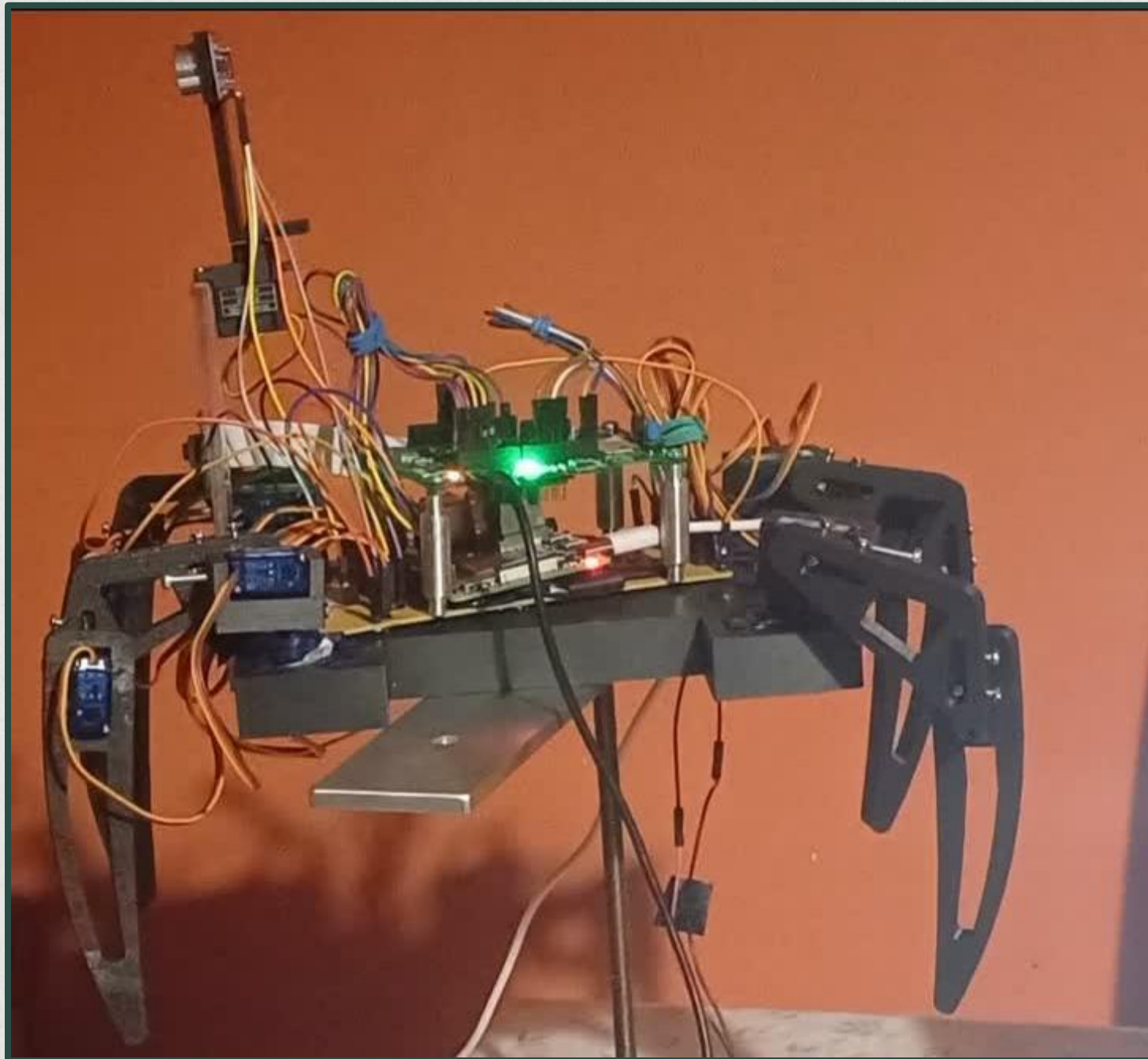
Alerting Search & Rescue

Motion JPEG (MJPEG) server

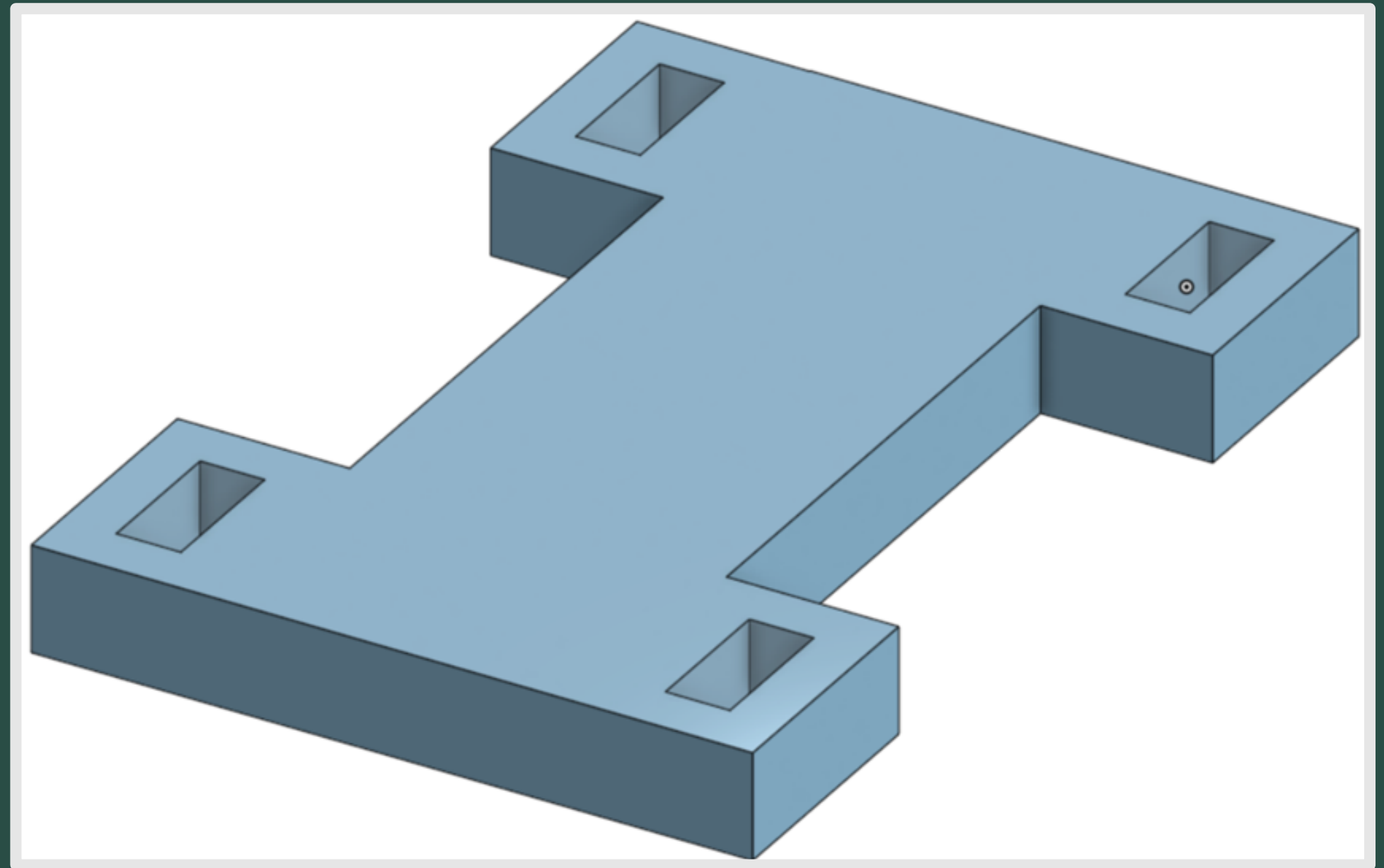
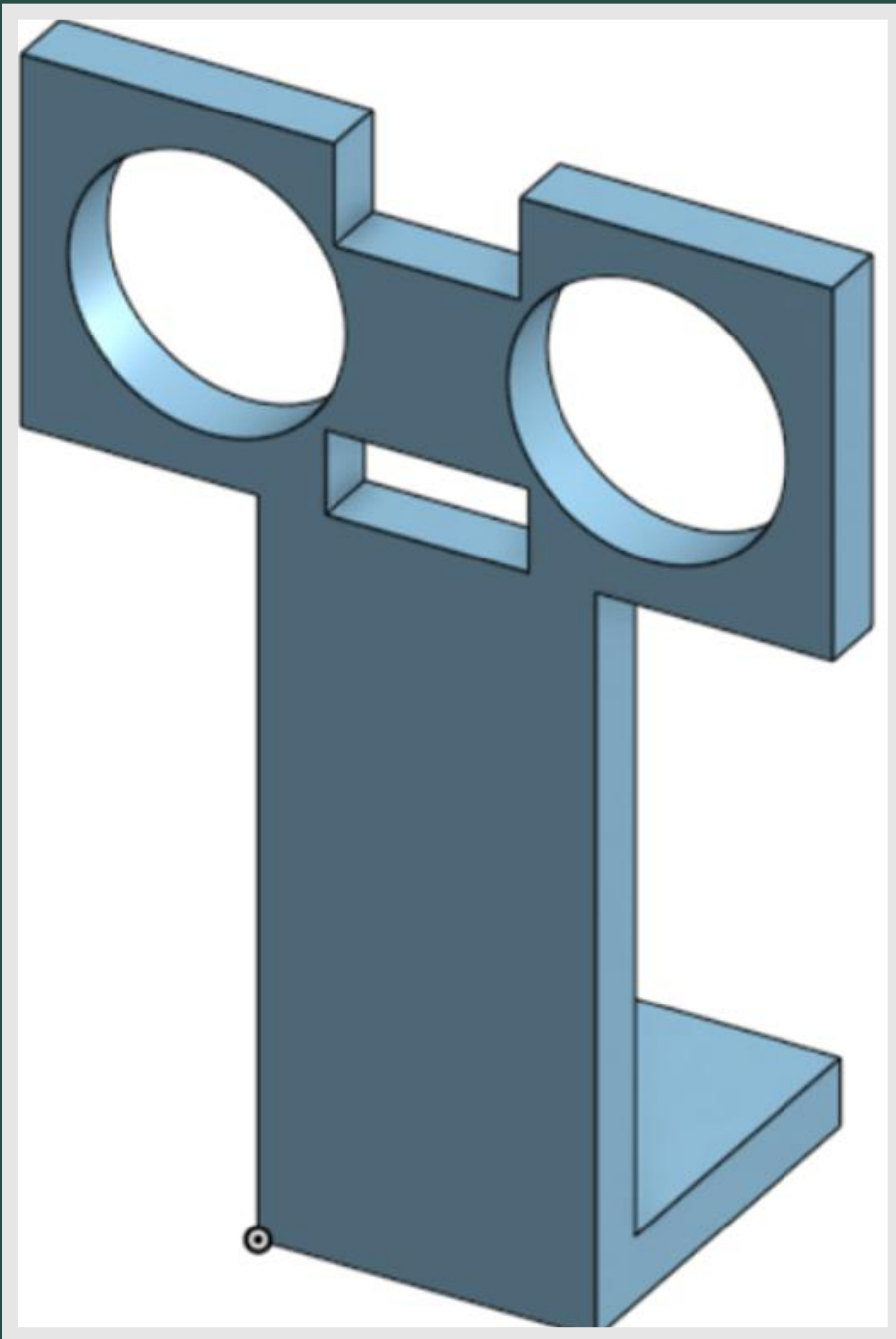
Simple mail transfer protocol server (SMTP)



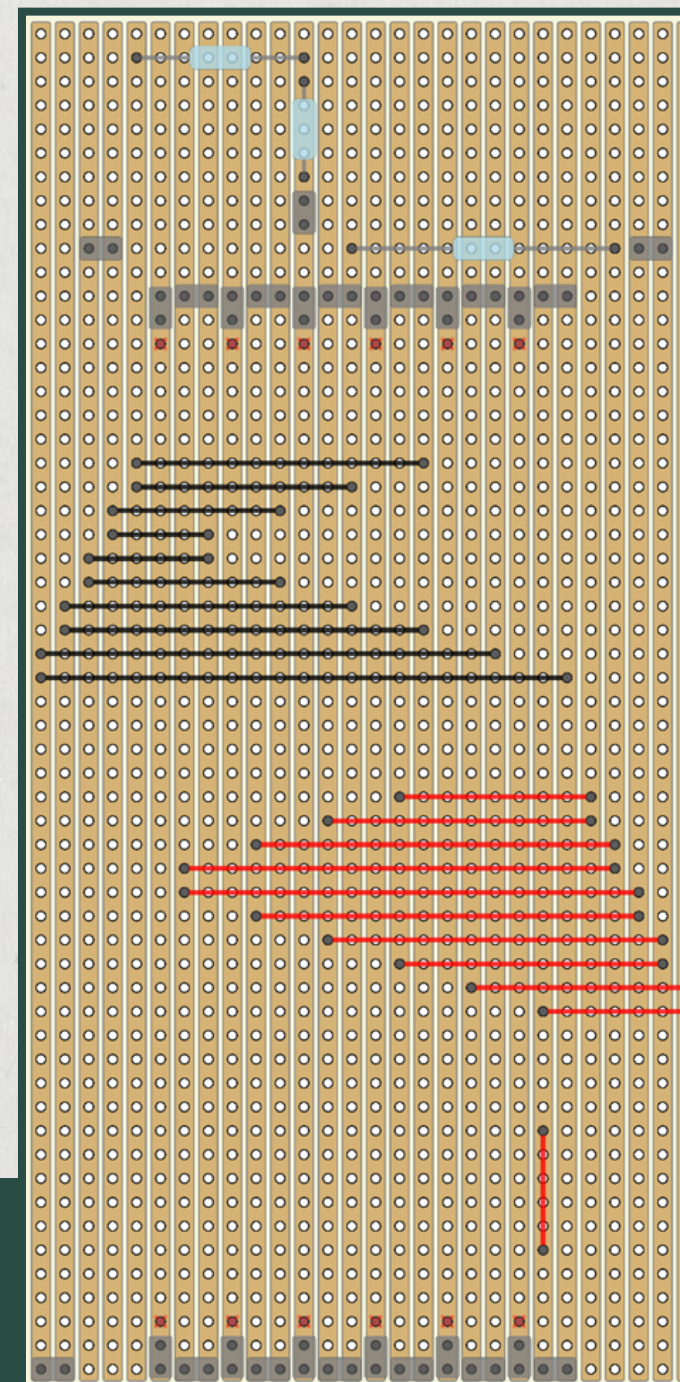
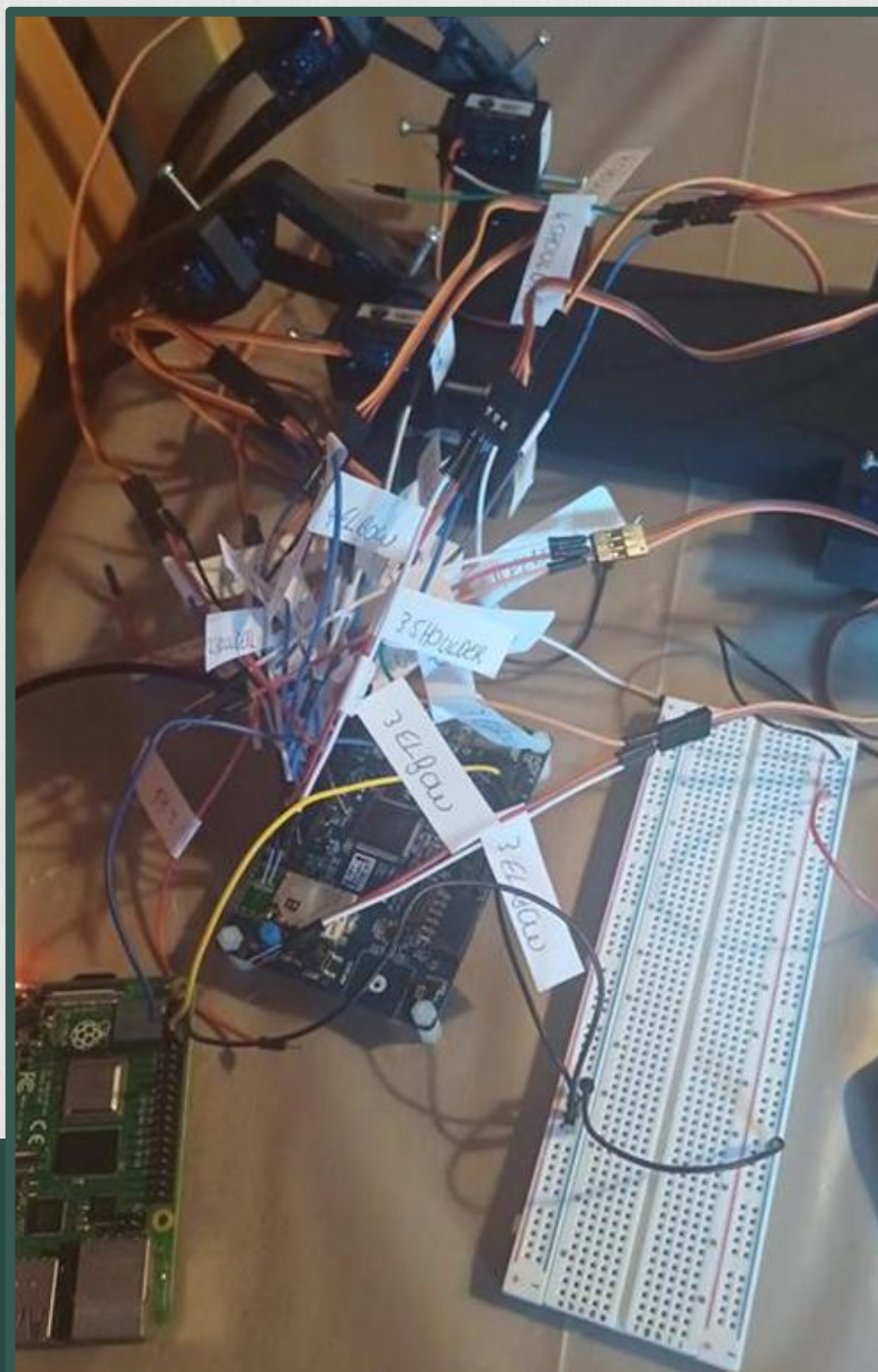
Detecting Collisions



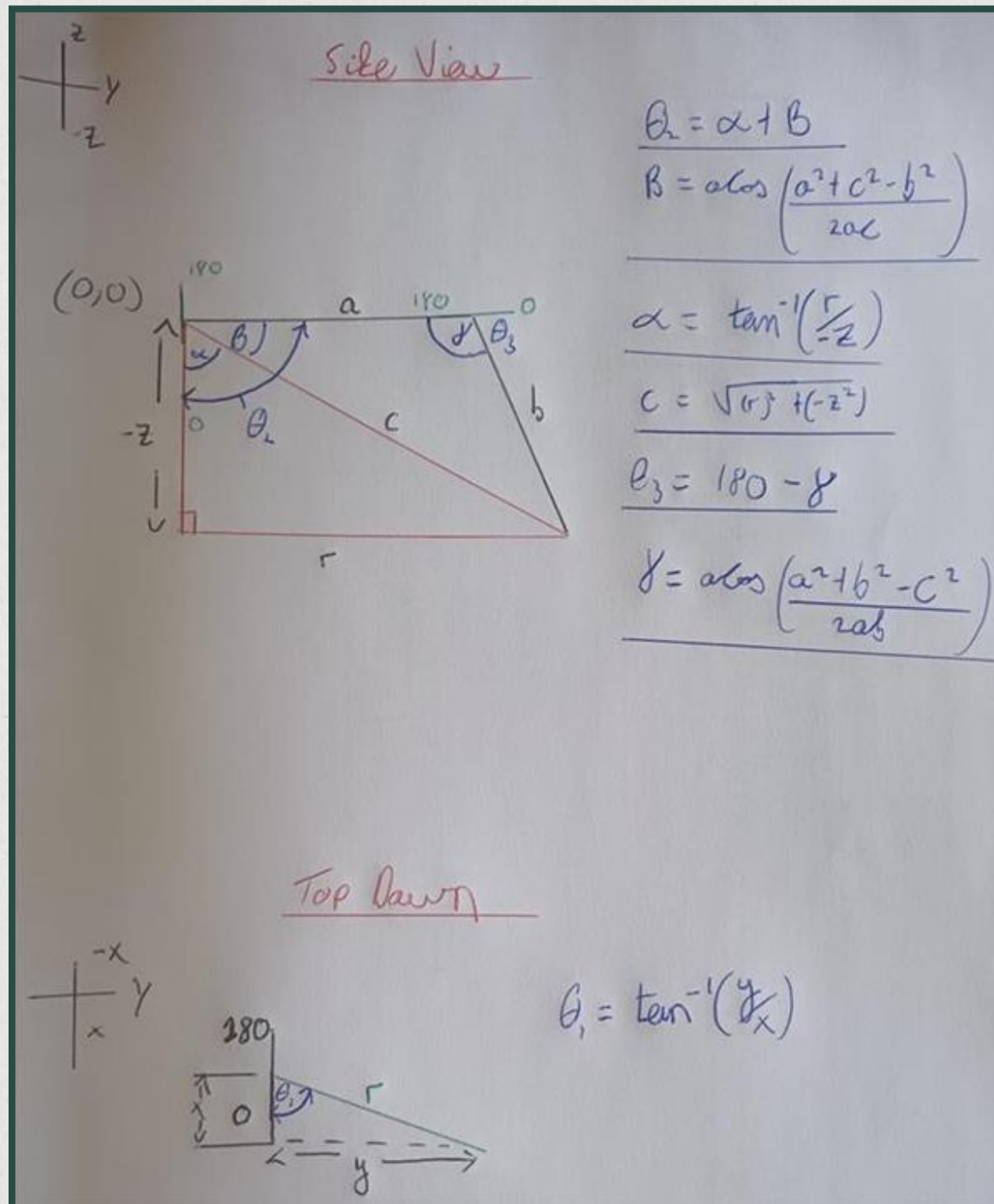
Building the SaRQ



Building the SaRQ



Inverse Kinematics (IK)

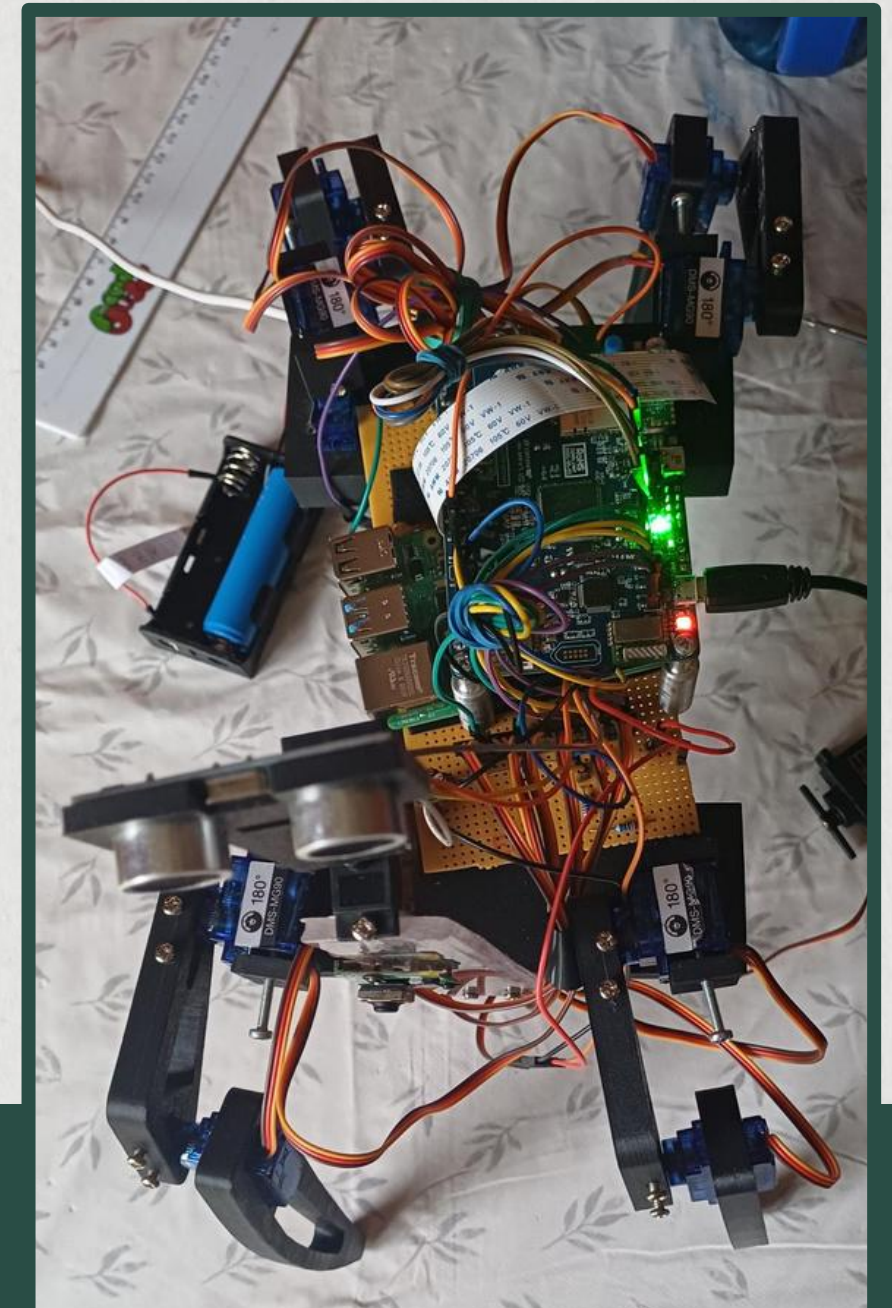
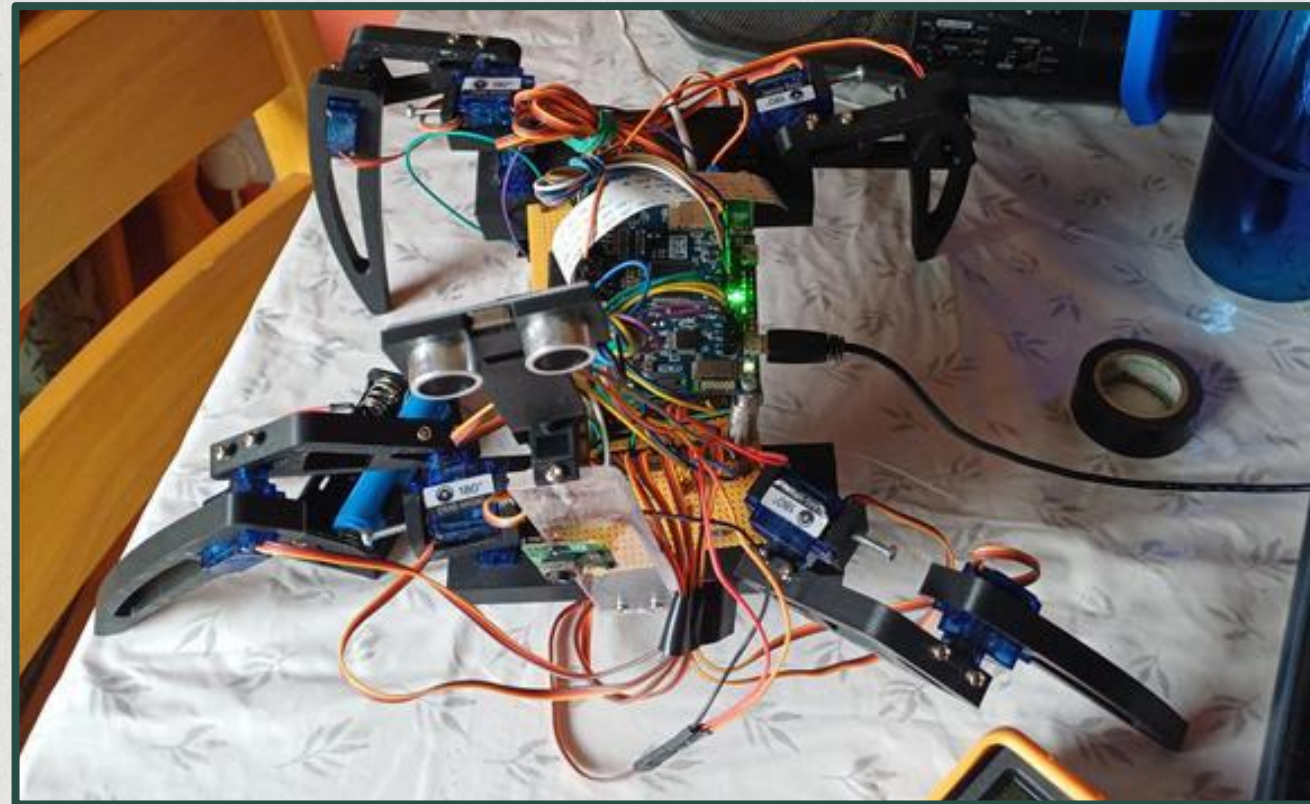
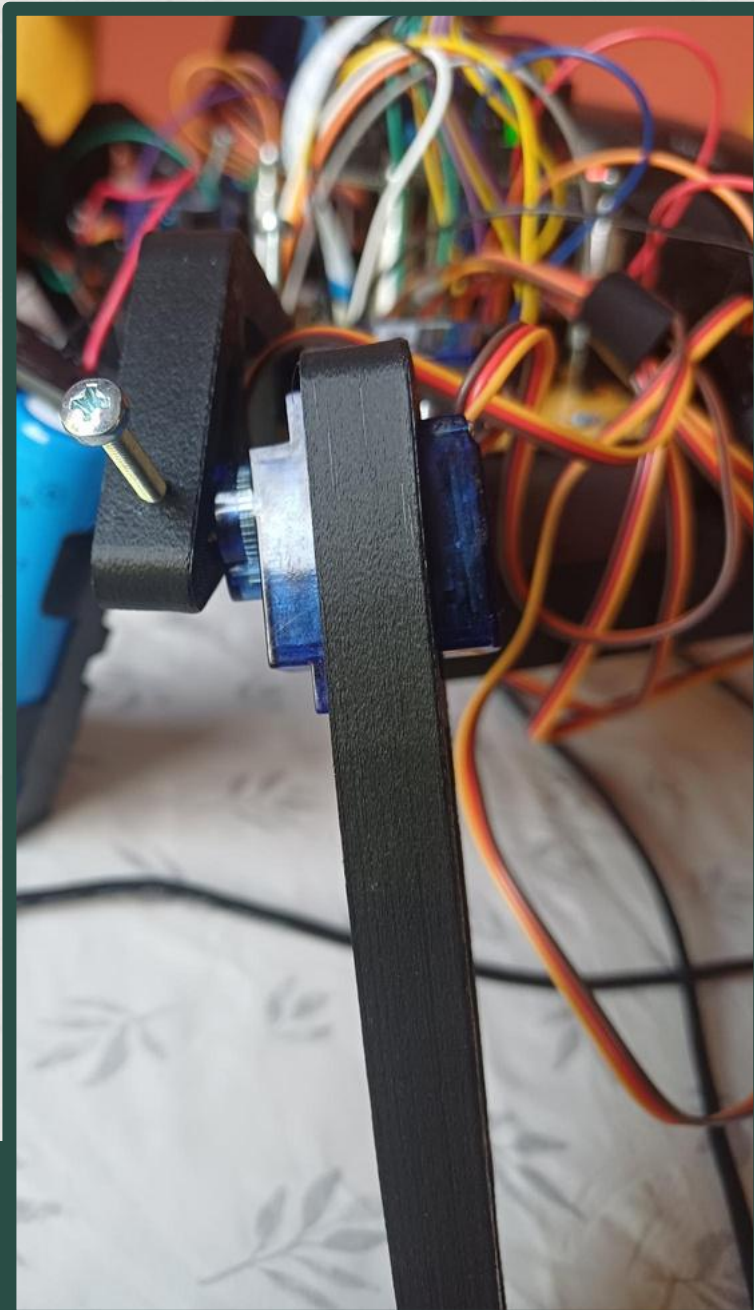


What is IK?

Trigonometric calculations

Tool for creating animations

Creating the Animation



Testing

YOLO testing

Facing Camera:

With movement: 0.86

Without movement: 0.75

Covered in "debris": 0.63

Back to Camera:

With movement: 0.77

Without movement: 0.65

Covered in "debris": 0.55

IK Model Testing

X-Axis

10mm – No deviation

39mm – Deviation in Y-Axis (10mm)

Y-Axis

10mm – No deviation

39mm – Deviation in X-Axis (5mm)

Project Management

[OneNote](#)

[Jira](#)

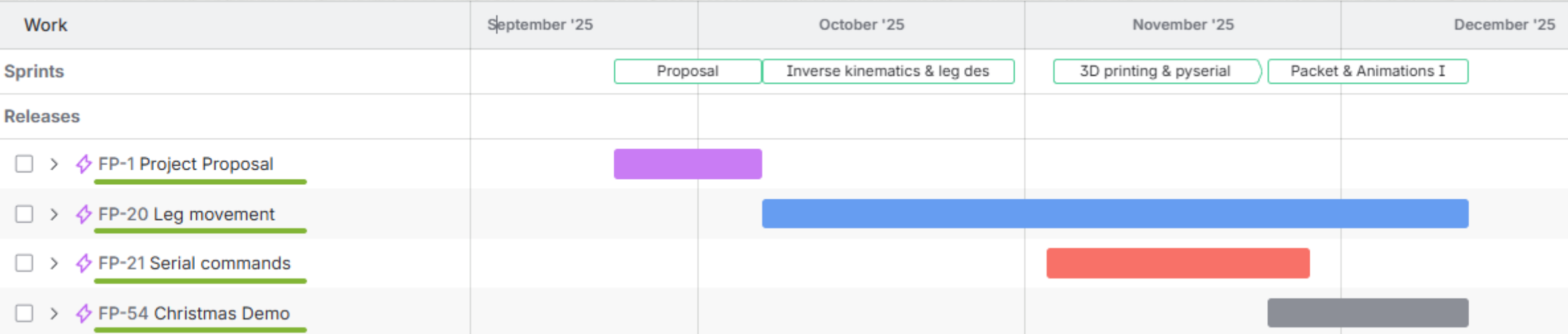
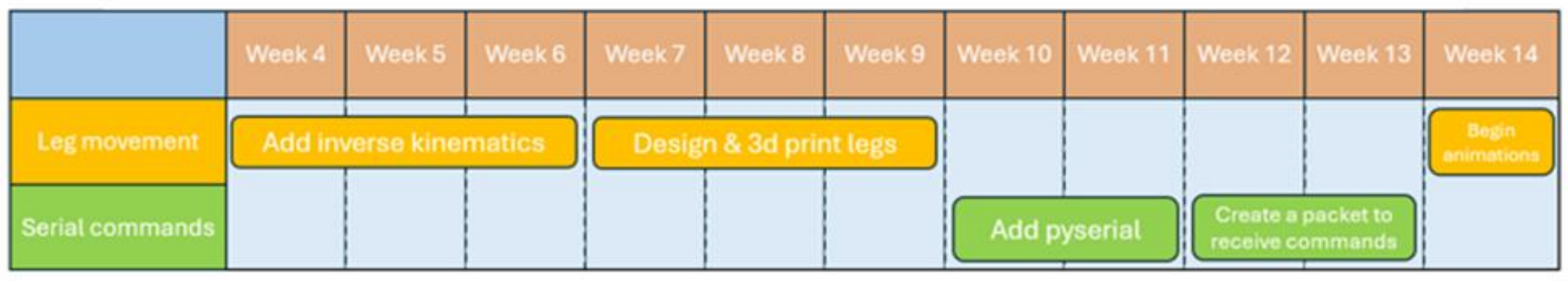
[GitHub](#)

Weekly stand-ups

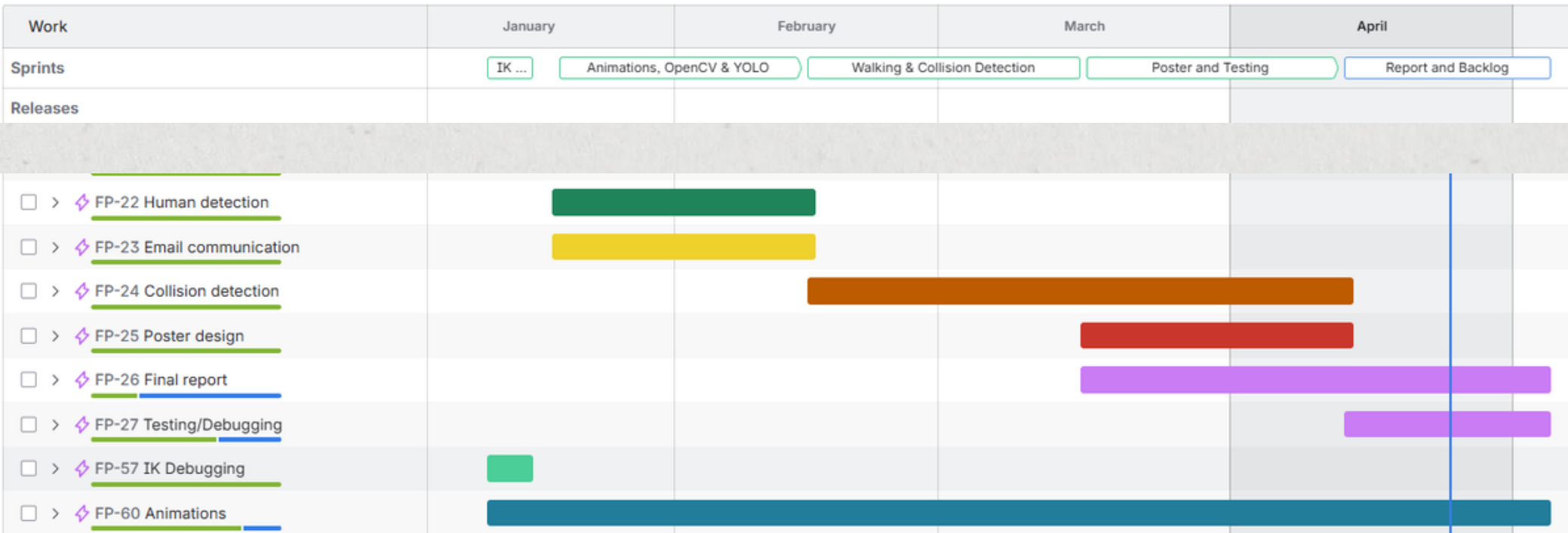
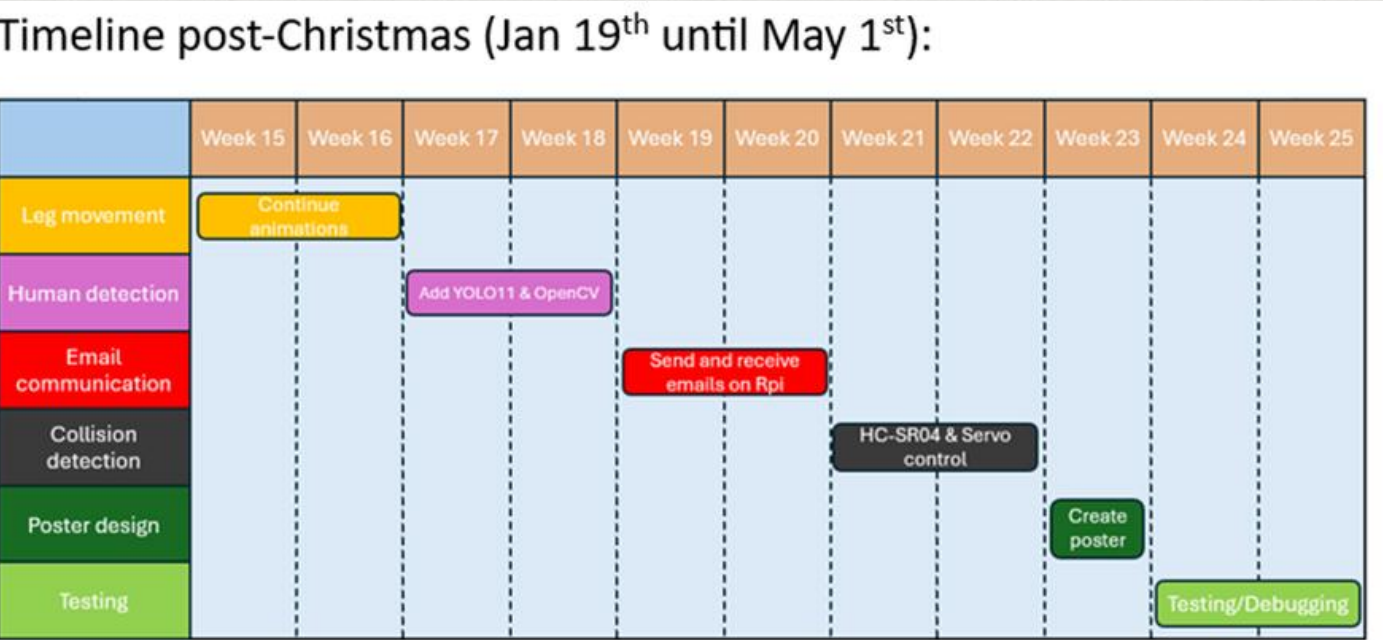


Project Timeline

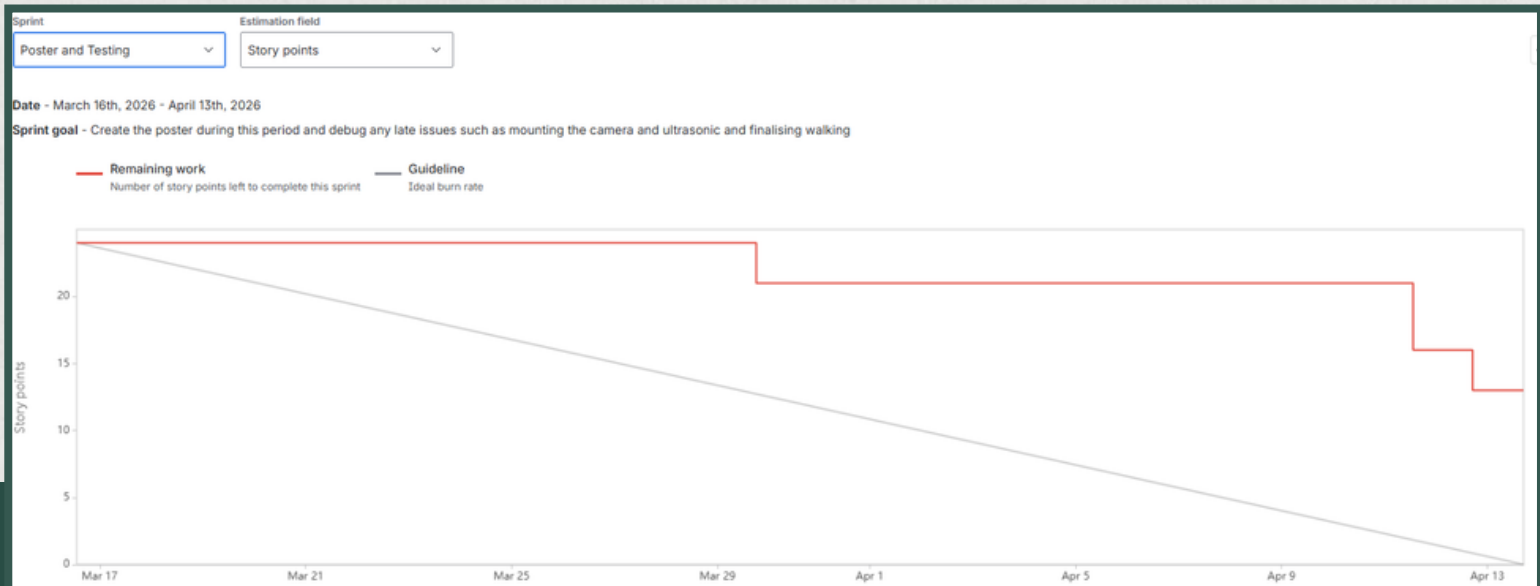
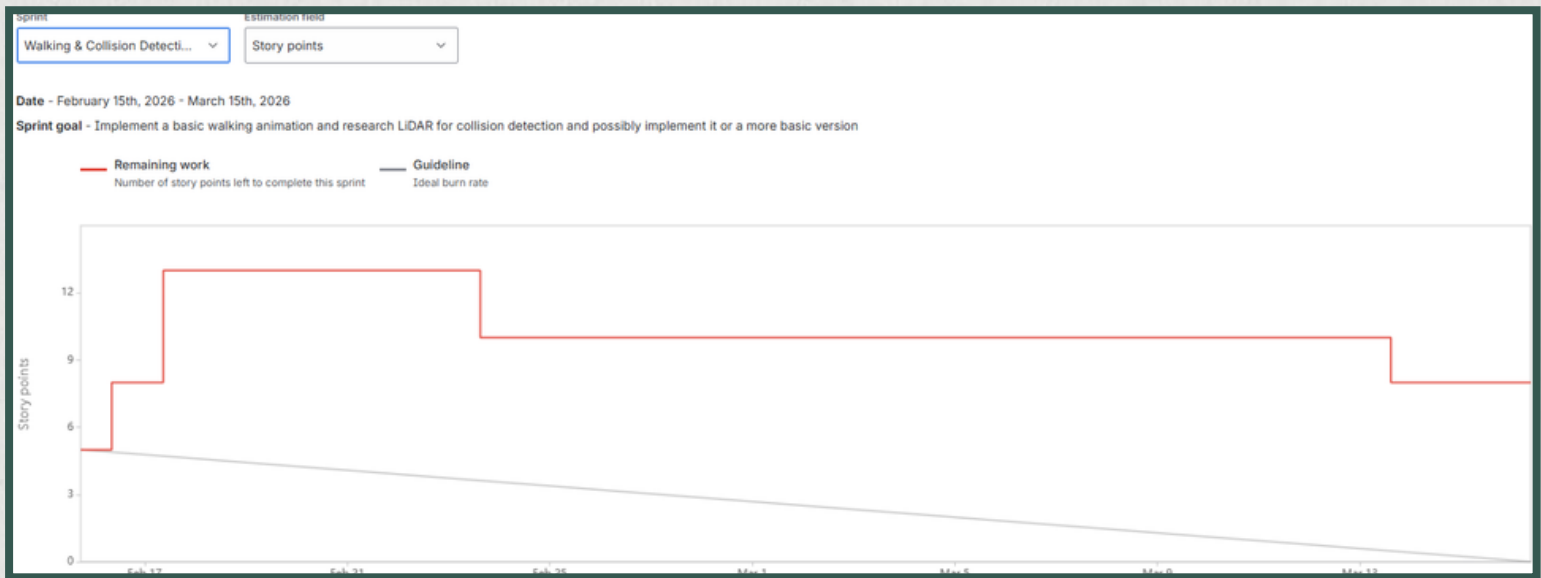
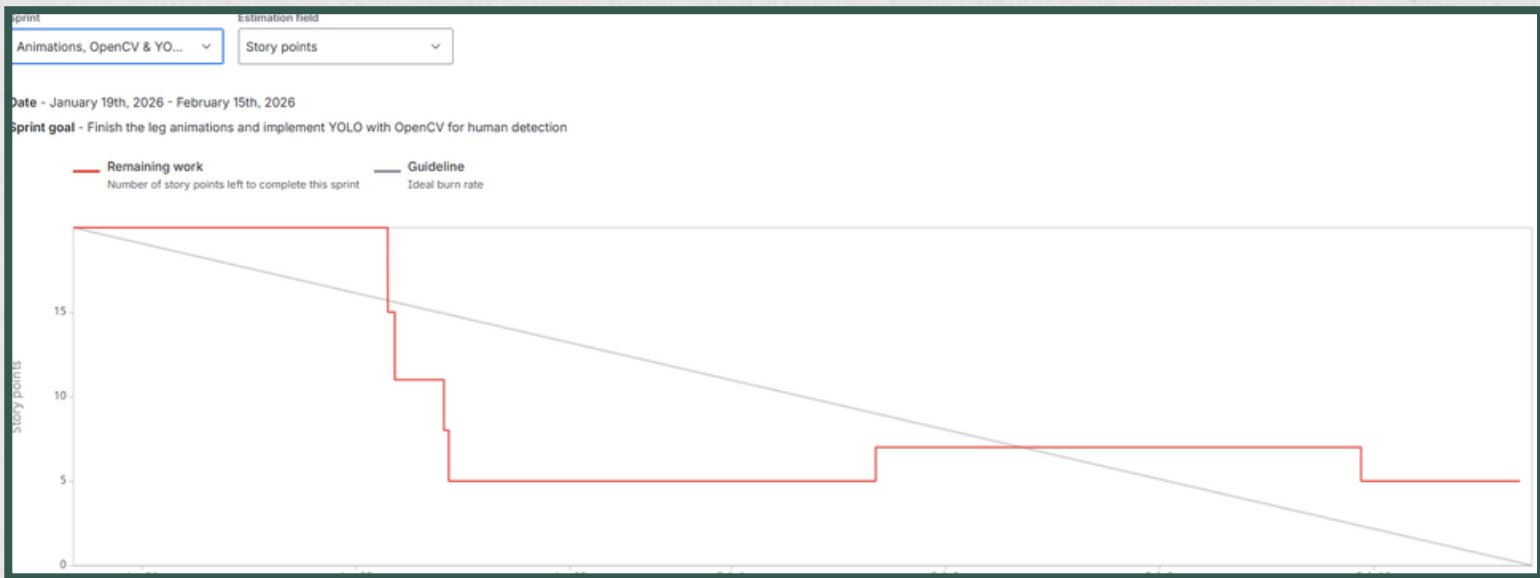
Timeline pre-Christmas (October 6th until December 21st):



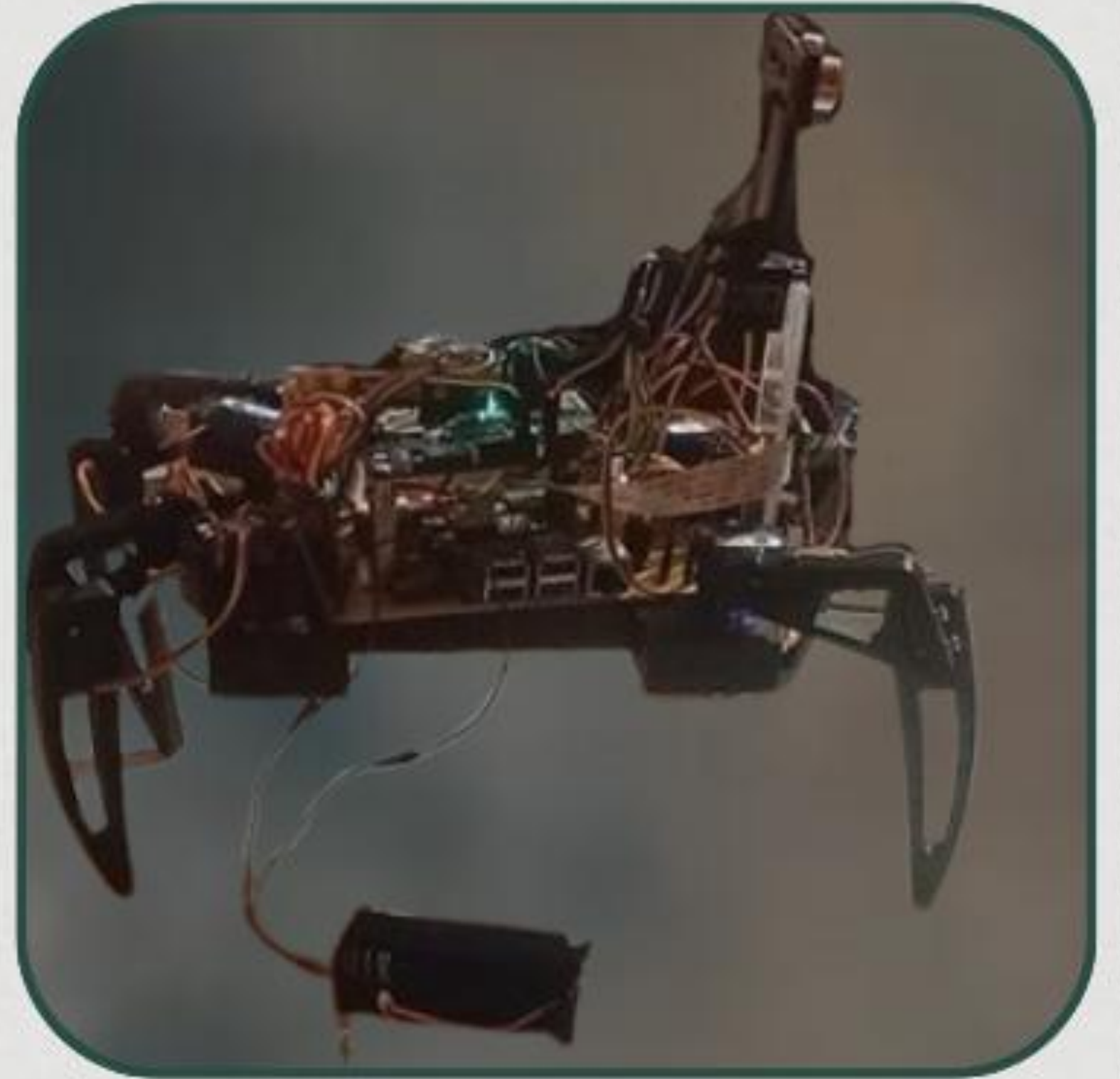
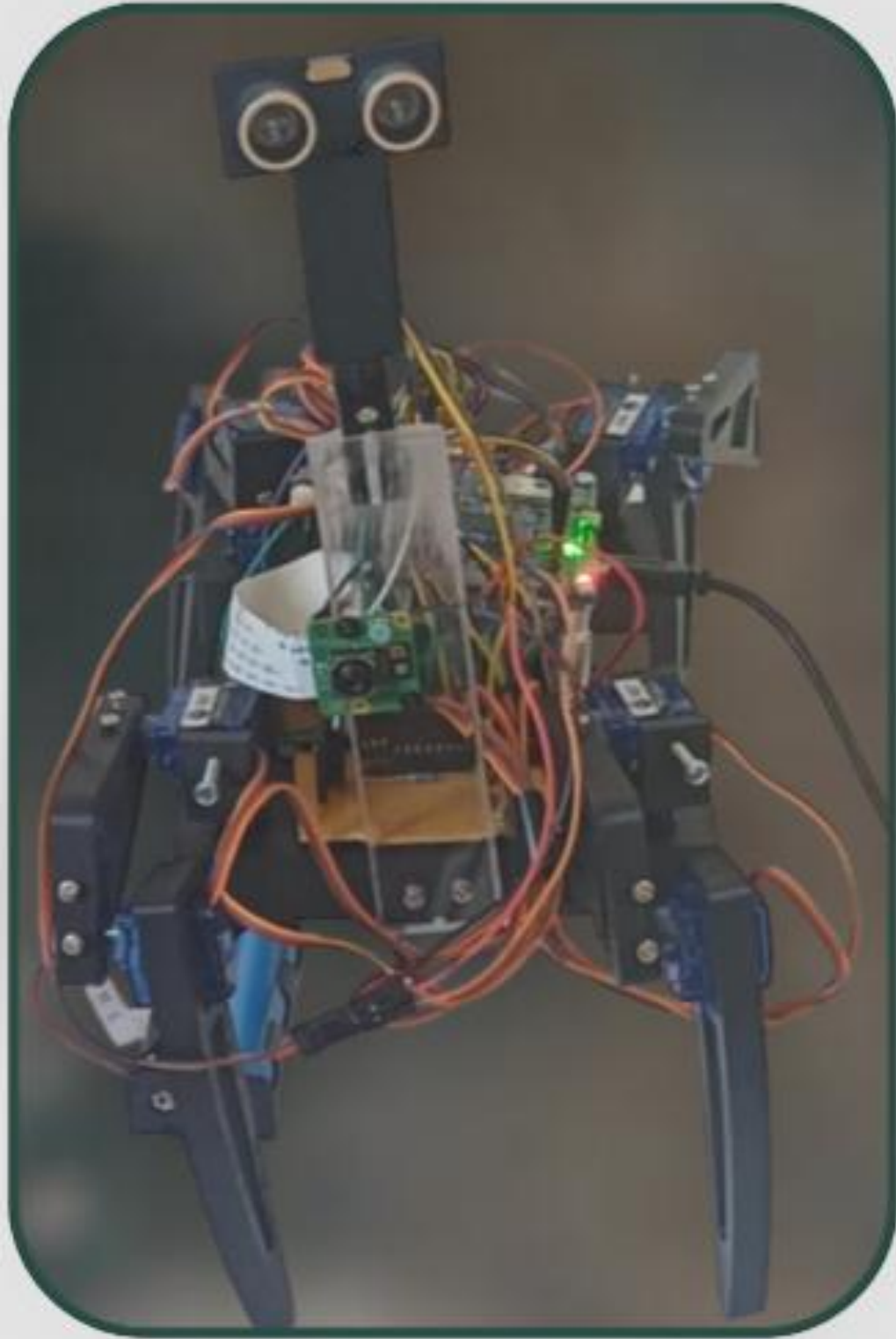
Project Timeline



Project Timeline



Final Product



Thank you
for listening